AI-Driven Smart Fashion Retail Recommendation System

Enhancing Customer Experience, Inventory Optimization, and Marketing Intelligence using AI



R26 – IT - 60

SLIIT
FACULTY OF COMPUTING

Demo


CapCut Ai
AI-Driven Smart Fashion
Retail Recommendation System

Problem & Impact

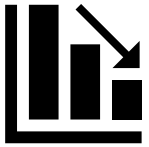
The Problem

- Traditional retail systems focus only on **sales and inventory**
- Retailers lack insights into:
  - customer behavior
  - product preferences
  - promotion effectiveness
- Large amounts of retail data remain **unused**
- **Absence on integrated systems**
- **Absence of smart systems for stakeholder decisions and use of manual traditional methods**



The Impact

- Poor customer experience
- Ineffective marketing strategies
- Stock shortages and overstock issues
- Reduced sales and business performance
- Making poor decisions based on manual analysis
- No clear view of customer behavior





Our Proposed System

GPS-Based Customer Behaviour Tracking & Staff Assistance

Tracks customer movement patterns and provides real-time assistance alerts to store staff.

AI-Based Virtual Try-On & Size Recommendation

Allows customers to virtually try products and receive accurate size recommendations using AI.

Smart Inventory & Stock Flow Optimization

Predicts product demand and optimizes inventory to prevent stock shortages and overstock.

Personalized Marketing Intelligence Dashboard

Analyzes customer data to generate targeted promotions and measure campaign performance.

IT22243812 | L.A.D.A.S. Perera

GPS-Based Customer Behaviour Tracking & Real-Time Staff Assistance



Problem Definition & Research Gap

Research Gap

Existing retail tracking systems mainly focus on customer monitoring using CCTV or sensors, with limited real-time staff assistance and privacy-friendly solutions.

Novelty

This research introduces a GPS-based zone tracking system that combines customer behaviour analysis, difficulty detection, real-time staff alerts, and store analytics in one solution.

Solution Overview

Problem

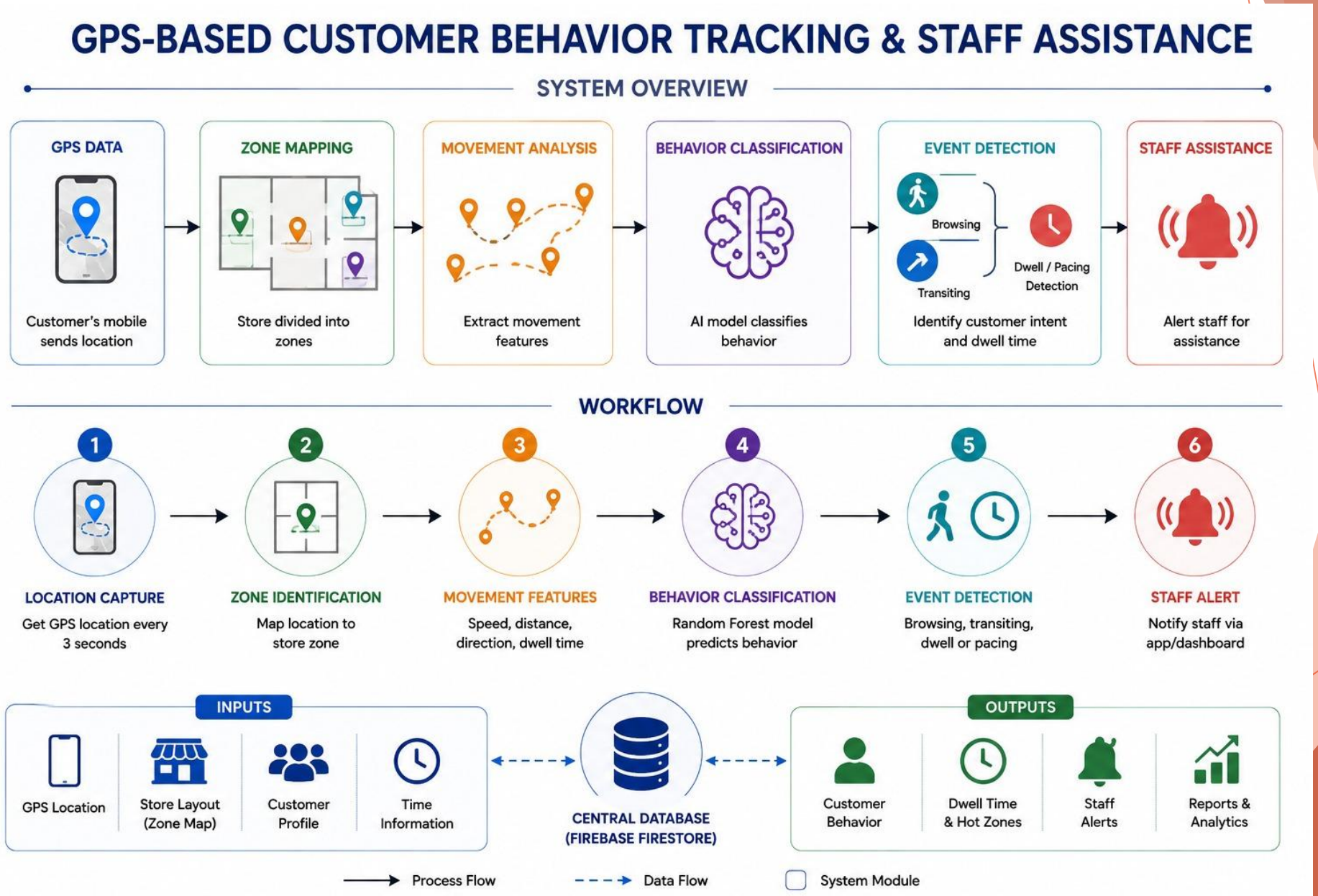
- Difficult to identify customers needing assistance
- Manual monitoring is inefficient
- Real-time observation is challenging

Solution

- GPS-based indoor customer tracking
- Detects confusion, hesitation, and product interest
- Helps staff provide timely assistance

GPS Based Customer Behavior Tracking and Staff Assistance

Work Flow



IT22284952 | Silva T. A

Specialization: Information Technology

AI-Based Virtual Try-On and Size Recommendation System

An AI-powered mobile application that facilitates virtual try-on



Problem Definition & Research Gap

Research Gap

Existing fashion systems rely on manual measurements or 3D scanning and lack accurate integration of virtual try-on with size prediction, leading to poor fit accuracy.

Novelty

Combines Gemini-based virtual try-on with ML-based size prediction to give accurate fit and visualization without 3D scanning or manual measurements.

Solution Overview

Problem

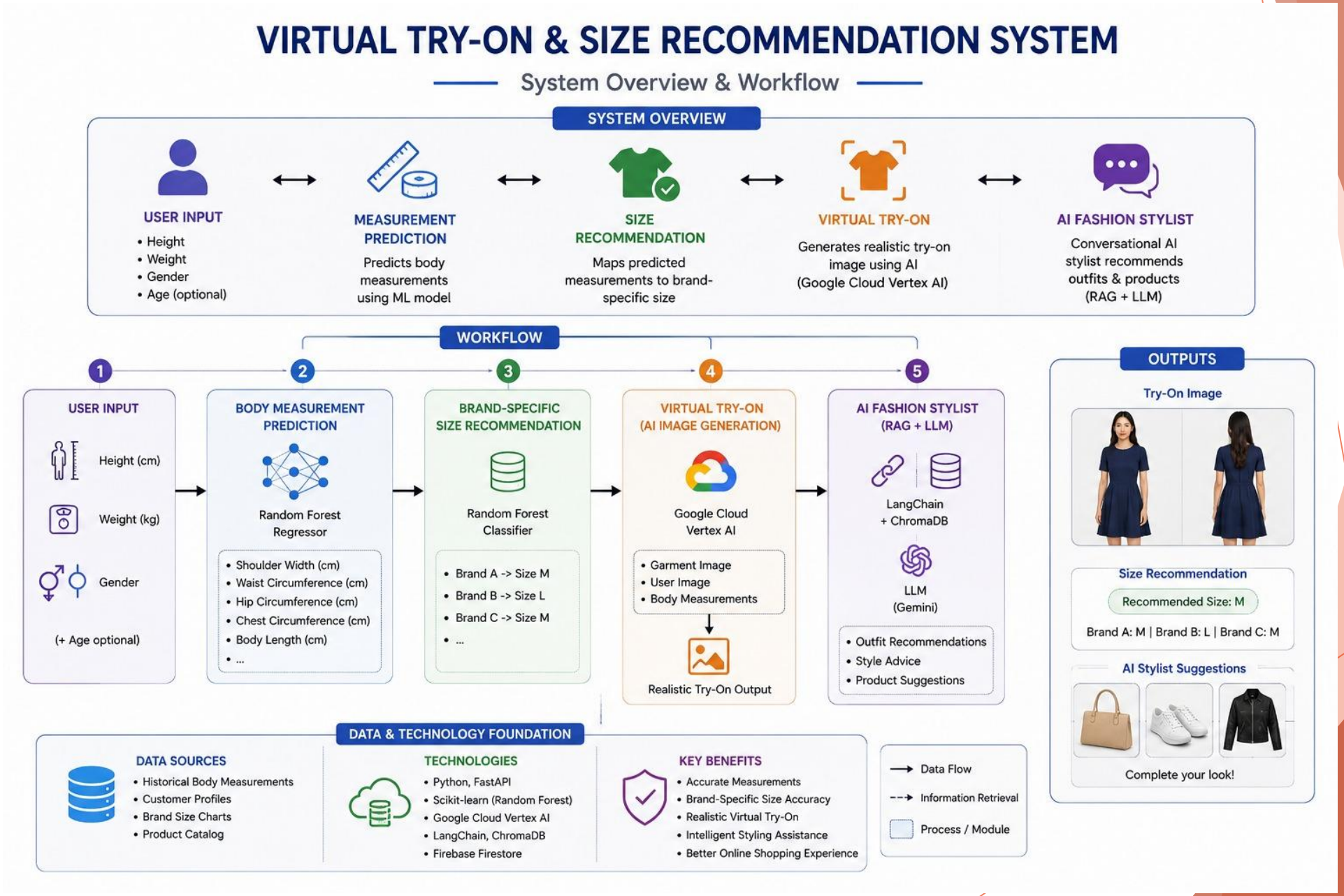
- Customers cannot try clothes online
- Wrong size selection issues

Solution

- AI generates virtual try-on results
- ML predicts accurate clothing sizes

Virtual Try On and Size Recommendation System

Work Flow



IT22244598 | Methsara W.M

Smart Inventory & Stock Flow Optimization System

AI-Powered Retail Inventory Intelligence
for
Omni-Retail Environments



Problem Definition & Research Gap

Research Gap

Existing retail inventory systems mainly focus on stock tracking and lack AI-based demand forecasting and intelligent inventory decision support.

Novelty

This research introduces an AI-powered inventory optimization system that combines demand forecasting, inventory analysis, and AI-generated business insights to improve retail stock management.

Solution Overview

Problem

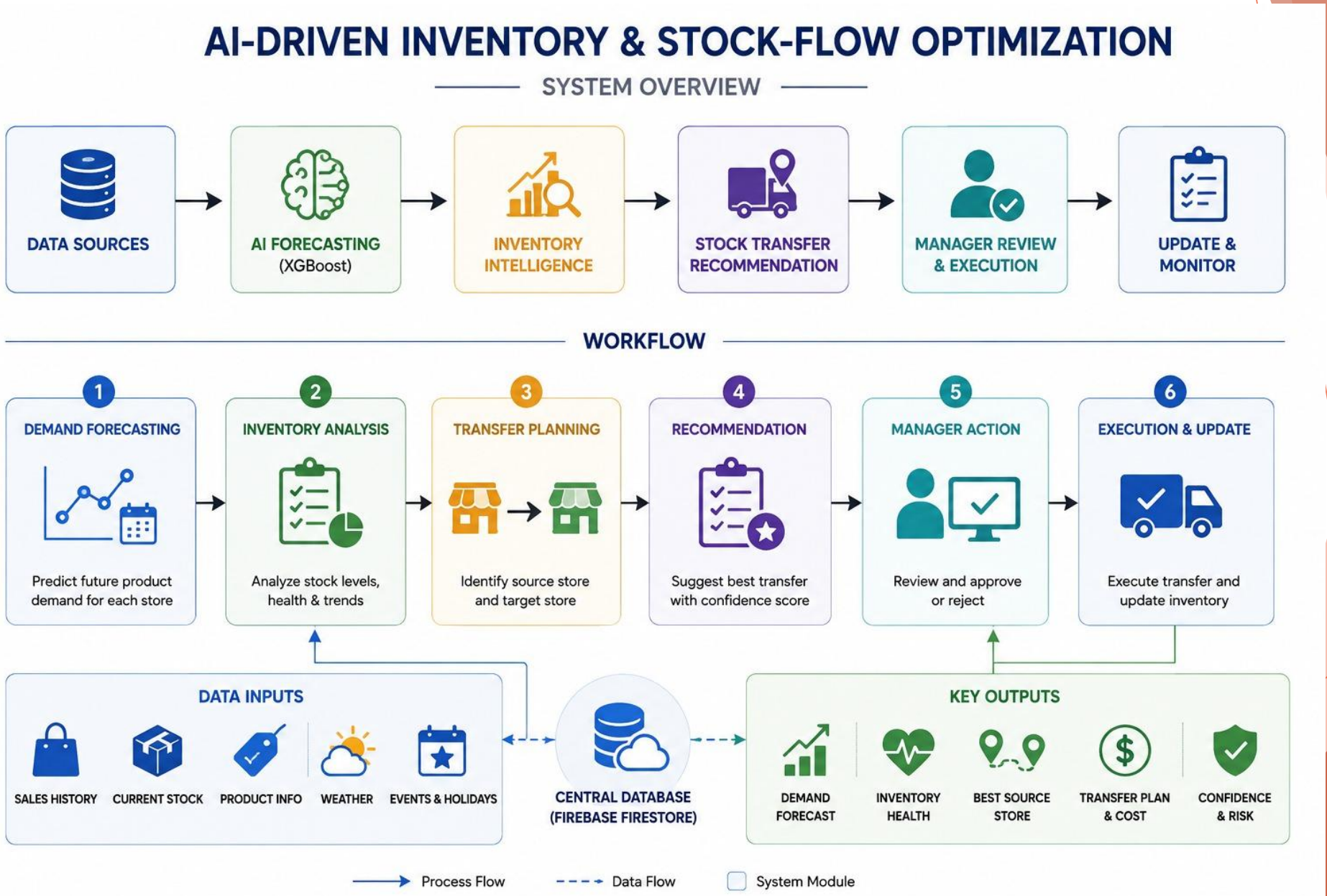
- Retail stores face stock shortages and overstock situations
- Limited intelligent inventory decision support
- Poor visibility of stock movement and demand trends
- Difficulty predicting future inventory demand accurately

Solution

- AI-based demand forecasting system
- Inventory optimization and stock recommendation engine
- AI-generated business insights and explanations
- Conversational AI retail assistant for inventory analysis

AI-Driven Inventory and Srock Flow Optimization

Work Flow



IT22146588 | WIJETUNGA J A R K

Personalized Marketing Intelligence

Contribution : Marketing Intelligence Dashboard

An intelligence marketing dashboard which analyses customer behaviors and help stakeholders to gain insight for better decision making

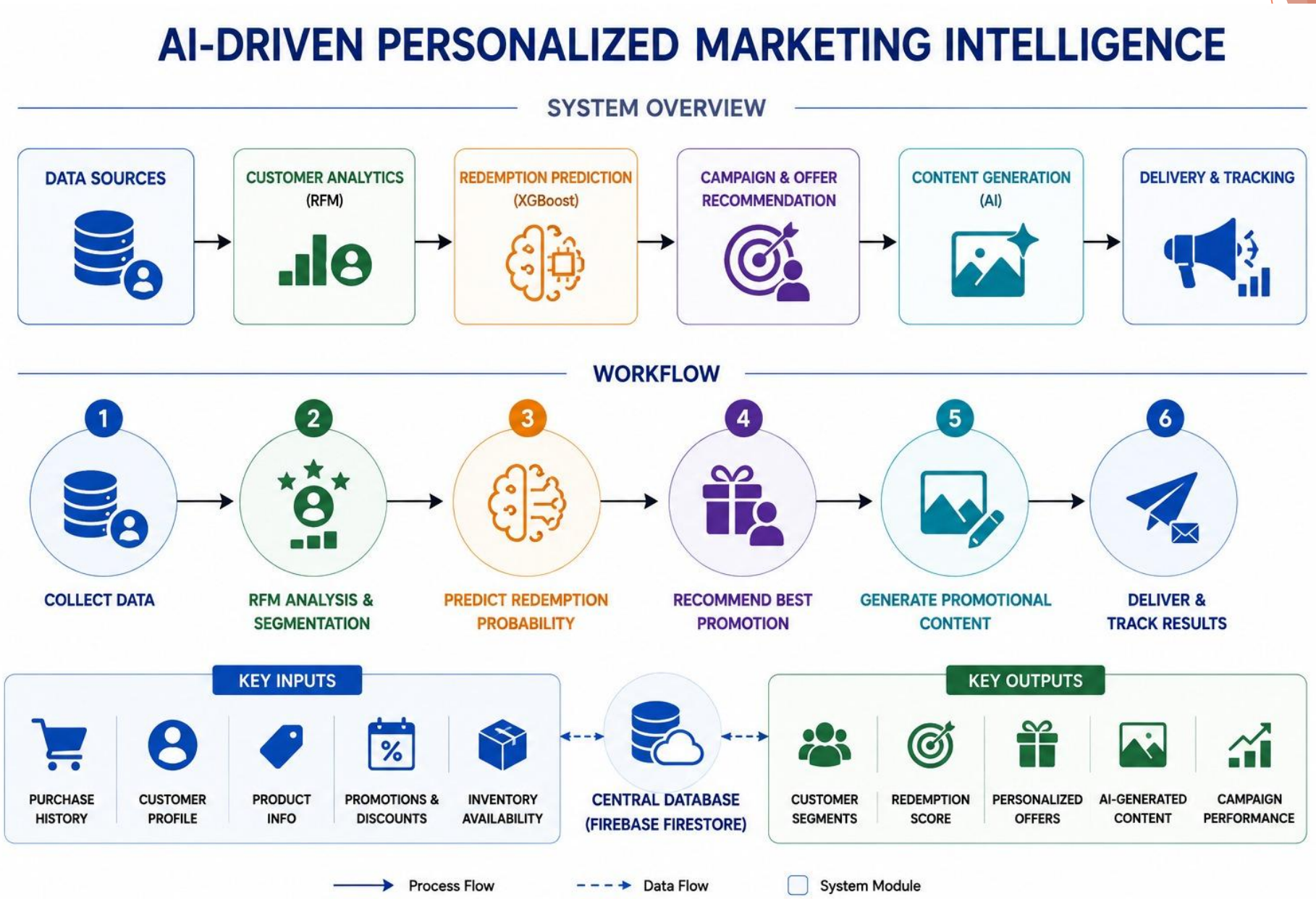


Problems identified and the proposed solutions (proto type)

No	Problem	Solution
1	Generic promotions	Recommender predicts best offer type per customer segment
2	Lack of insights into promotion performance and customer responses	Promotion Performance tab shows CTR, redemption rates, revenue lift
3	Low customer engagement due to poorly targeted promotions	Customized poster generator creates targeted visuals per age/gender
4	Limited understanding of individual customer preferences and purchasing behavior	Promo Attributes tab analyses preferences by age group, gender
5	Inability to identify hidden or future customer demand	Predictions which promotions customers will respond to before campaigns run
6	Manual marketing campaign creation which reduces efficiency	Automated poster generation and batch segment recommendations reduce manual effort

AI Driven Personalized Marketing Intelligence

Work Flow



Future Work

- Improve demand forecasting using hybrid ML models and additional data sources (e.g., live traffic, POS data)
- Expand virtual try-on with larger training datasets and fine-tuned generative models
- Explore alternative indoor-positioning technologies alongside GPS
- Enable adaptive, continuously learning models across all four platform components
- Conduct full-platform deployment and real-world testing to validate effectiveness and scalability

